

```
String reversed = new StringBuilder(str).reverse().toString();
```

Better Answer

In production I'd use StringBuilder because it's simple and optimized. If the interviewer wants logic implementation, I can do it using a loop.

Manual:

```
String rev = "";

for(int i=str.length()-1;i>=0;i--){
    rev += str.charAt(i);
}
```

2. Palindrome

Question

```
String str = "madam";
```

Check palindrome.

Answer

```
String rev = new StringBuilder(str)
    .reverse()
    .toString();

System.out.println(str.equals(rev));
```

Better

```
int left=0;
int right=str.length()-1;

while(left<right){
    if(str.charAt(left)!=str.charAt(right)){
        return false;
    }
    left++;
    right--;
}
return true;
```

Mention:

Two-pointer approach avoids creating extra string.

3. Find Duplicate Elements

Input

```
int arr[] = {1,2,3,2,4,5,1};
```

Answer

```
Set<Integer> set = new HashSet<>();

for(int num : arr){
    if(!set.add(num)){
        System.out.println(num);
    }
}
```

Interview Point

HashSet lookup is $O(1)$, overall complexity $O(n)$.

4. Count Frequency

Question

Count character frequency.

```
hello
```

Answer

```
Map<Character,Integer> map = new HashMap<>();

for(char ch : str.toCharArray()){
    map.put(ch,map.getDefault(ch,0)+1);
}

System.out.println(map);
```

Output

```
h=1
e=1
l=2
o=1
```

5. Remove Duplicates From List

Input

```
[1,2,3,1,2,4]
```

Answer

```
List<Integer> list =  
Arrays.asList(1,2,3,1,2,4);  
  
List<Integer> result =  
new ArrayList<>(new LinkedHashSet<>(list));
```

Why LinkedHashSet?

Removes duplicates and preserves insertion order.

6. Second Largest Number

Question

```
int arr[]={10,20,5,40,30};
```

Answer

```
int largest=Integer.MIN_VALUE;  
int second=Integer.MIN_VALUE;  
  
for(int num:arr){  
  
    if(num>largest){  
        second=largest;  
        largest=num;  
    }  
    else if(num>second && num!=largest){  
        second=num;  
    }  
}  
  
System.out.println(second);
```

Output

7. Java 8 Even Numbers

Question

Find even numbers.

Answer

```
List<Integer> even =  
list.stream()  
    .filter(n -> n % 2 == 0)  
    .collect(Collectors.toList());
```

Follow-up

Difference between map and filter?

```
filter -> removes elements
```

```
map -> transforms elements
```

8. Sort Employees By Salary

Question

Sort Employee list.

Answer

```
employees.sort(  
    Comparator.comparing(Employee::getSalary)  
);
```

Descending

```
employees.sort(  
    Comparator.comparing(Employee::getSalary)  
                .reversed()  
);
```

9. HashMap vs ConcurrentHashMap

Interviewer

Difference?

Answer

HashMap is not thread-safe. ConcurrentHashMap is thread-safe and supports concurrent reads and writes with better performance than Hashtable.

10. Output Question

Question

```
String s1="hello";  
String s2="hello";  
  
System.out.println(s1==s2);
```

Output

true

Reason

Both point to same String Pool object.

Question

```
String s1=new String("hello");  
String s2=new String("hello");  
  
System.out.println(s1==s2);
```

Output

```
false
```

Why?

Different objects.

Question

```
System.out.println(s1.equals(s2));
```

Output

```
true
```

Because content comparison.

11. Exception Question

Question

Output?

```
try{
    int x=10/0;
}
catch(Exception e){
    System.out.println("Caught");
}
finally{
    System.out.println("Finally");
}
```

Output

```
Caught
Finally
```

Interview Point

finally executes whether exception occurs or not.

12. Custom Exception

Answer

```
public class UserNotFoundException
extends RuntimeException{

    public UserNotFoundException(String msg){
        super(msg);
    }
}
```

Throw:


```
throw new UserNotFoundException(  
    "User not found"  
);
```

SQL 1 : Second Highest Salary

Table

EMPLOYEE	
id	salary
1	1000
2	5000
3	3000

Answer

```
SELECT MAX(salary)  
FROM employee  
WHERE salary <  
(  
    SELECT MAX(salary)  
    FROM employee  
);
```

SQL 2 : Nth Highest Salary

Answer

```
SELECT salary
FROM
(
  SELECT salary,
  DENSE_RANK()
  OVER(ORDER BY salary DESC) rnk
  FROM employee
)t
WHERE rnk=3;
```

SQL 3 : Duplicate Records

Question

Find duplicate emails.

Answer

```
SELECT email,
COUNT(*)
FROM employee
GROUP BY email
HAVING COUNT(*) > 1;
```

SQL 4 : Department Wise Highest Salary

Answer

```
SELECT *
FROM
(
  SELECT e.*,
  ROW_NUMBER()
  OVER
  (
```

```
    PARTITION BY dept
    ORDER BY salary DESC
) rn
FROM employee e
)t
WHERE rn=1;
```

SQL 5 : INNER JOIN vs LEFT JOIN

Interview Answer

INNER JOIN

Returns only matching records.

LEFT JOIN

Returns all rows from left table and matching rows from right table.

This is asked in almost every Java interview.

SQL 6 : Employees Joined Last 30 Days

```
SELECT *
FROM employee
WHERE joining_date >=
CURRENT_DATE - 30;
```

One Very Common Java 8 Question

Question

Find duplicate elements using Streams.

```
List<Integer> list =  
Arrays.asList(1,2,3,2,4,1);
```

Answer

Java

```
Set<Integer> set = new HashSet<>();  
  
list.stream()  
    .filter(n -> !set.add(n))  
    .forEach(System.out::println);
```

Output

Java

```
2  
1
```

For a 3–4 year Java Full Stack interview, if you master these 15–20 snippet questions plus Collections, Streams, Exception Handling, and SQL window functions, you'll cover about **80% of the coding questions** typically asked in service-based and product-based company interviews.

